

★ WEEK 14 · PART 2 · PRACTICE

Metropolis and Gibbs, Worked by Hand

Take One Step, Tune the Proposal, or Skip the Rejection

The acceptance ratio and the sampler's one dial

Part 2's samplers, with the arithmetic shown. We take one Metropolis-Hastings step - watching the evidence cancel and a worse move get accepted - then tune the proposal by its acceptance rate, and see how Gibbs skips rejection entirely.

2P

PART 2 · APPLICATIONS

Propose, Accept, or Tune

Case A: one Metropolis-Hastings step - the acceptance ratio, the evidence cancelling, and why a worse move is sometimes taken. Case B: the acceptance-rate dial that tunes the proposal, and Gibbs as the no-rejection alternative. Then a Claude Code lab.

What Part 2 Told Us

Now we take a step - with numbers

1

Propose a move, then decide

Metropolis-Hastings proposes a candidate and accepts it with a probability set by one ratio of target densities.

2

The evidence cancels

Because the ratio divides two posteriors, the intractable normalizer drops out - you never compute the hard integral.

3

Tune the proposal, or use Gibbs

The proposal step size sets the acceptance rate, which you tune. Gibbs instead draws each conditional exactly - no rejection.

Everything here is a calculator exercise: one acceptance ratio, three step sizes, one conditional.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from a current state or a step size.
- Compute the ratio, or the acceptance rate.
- Then read the accept, reject, or tune.

What you should watch

- How the evidence cancels in the ratio.
- How a worse move can still be accepted.
- How the step size sets efficiency.

One Metropolis-Hastings Step

Propose, score, accept or reject

Sampling the posterior of a strategy's expected return, we sit at 8% and propose two moves. Each is scored by the ratio of unnormalized posteriors - likelihood times prior - so the intractable evidence cancels. An uphill move is always taken; a downhill one is accepted with probability equal to the ratio. We walk both by hand

The current state

- At $\theta = 8\%$.
- Unnormalized target: $0.50 \times 0.60 = 0.30$.
- Likelihood times prior.

What we'll do by hand

- Score an uphill proposal to 11%.
- Score a downhill proposal to 4%.
- Decide each with a uniform draw.

The Acceptance Rule

One ratio, and the evidence cancels

$$a = \min\left(1, \frac{p(\theta'|D)}{p(\theta|D)}\right) = \min\left(1, \frac{L(\theta')\pi(\theta')/P(D)}{L(\theta)\pi(\theta)/P(D)}\right) \quad (\text{the evidence cancels})$$

The ratio

- Divide the proposed target by the current.
- Both are likelihood times prior.
- Accept with that ratio, capped at one.

Why it's magic

- The evidence sits in both, top and bottom.
- So it cancels - never computed.
- That's what makes MCMC possible.

An Uphill Move

Higher target, always accepted

$$\text{uphill } \theta=8\% \rightarrow \theta'=11\% : \text{ ratio} = \frac{0.75(0.56)}{0.50(0.60)} = \frac{0.42}{0.30} = \mathbf{1.40} > 1 \Rightarrow \text{ACCEPT}$$

The calculation

- Proposed target: 0.75 times 0.56 = 0.42.
- Divide by the current 0.30.
- The ratio is 1.40.

The decision

- A ratio above one means always accept.
- The move improves the target.
- The chain steps to 11%.

A Downhill Move

Sometimes taken anyway

$$\text{downhill } \theta''=4\% : \text{ratio} = \frac{0.18}{0.30} = \mathbf{0.60}$$

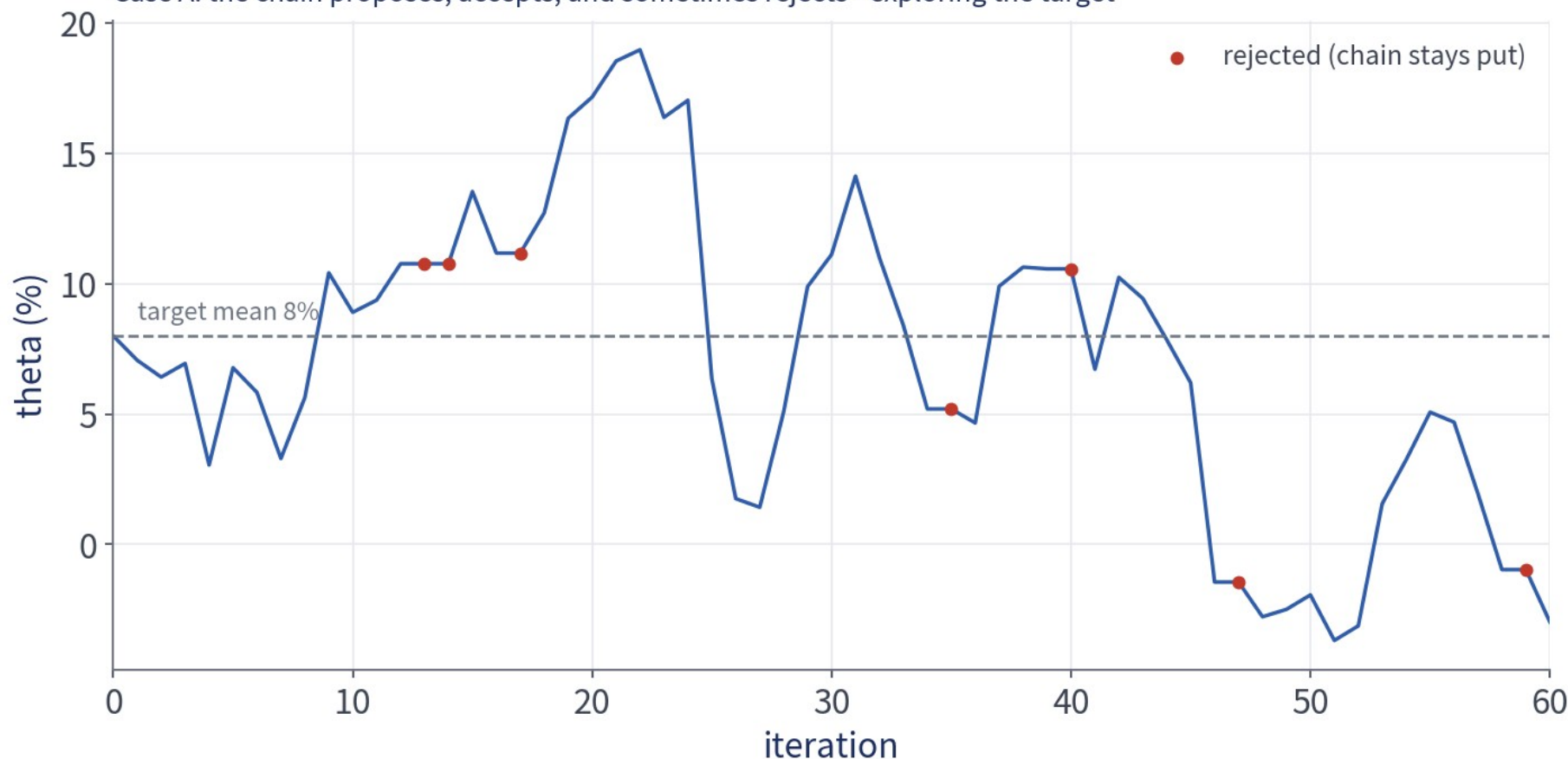
draw $u = 0.45 < 0.60 \Rightarrow$ ACCEPT (a worse move, for exploration)

The downhill move to 4% scores a ratio of 0.60, so it's accepted 60% of the time. A uniform draw of 0.45 falls below 0.60, so the chain takes the worse move - that's exploration.

Proposing, Accepting, Rejecting

The chain wanders around the target's center of 8%, taking most uphill moves and some downhill ones. The red points mark rejected proposals, where it stays put. Over many steps this exploration traces out the whole posterior - the accepted states are the samples

Case A: the chain proposes, accepts, and sometimes rejects - exploring the target



Why Accept Worse Moves

Exploration beats greed

If it only climbed

- The chain would rush to the peak and stop.
- It would never see the tails.
- The samples would misrepresent the posterior.

By taking some losses

- It explores the whole distribution.
- It visits regions in proportion to their mass.
- That's what makes the samples valid.

Tune the Proposal, or Skip It

The acceptance dial and Gibbs

Metropolis-Hastings has one real knob: the proposal step size, which sets the acceptance rate. Too small and it crawls, accepting almost everything but barely moving; too large and it stalls, rejecting almost everything. The sweet spot is in between. Gibbs avoids the whole issue by drawing each conditional exactly

The dial

- Small steps: acceptance near 0.90, tiny moves.
- Large steps: acceptance near 0.06, mostly stuck.
- The target is roughly a third.

What we'll do by hand

- Read the acceptance rate at three step sizes.
- Identify the efficient one.
- Contrast with a Gibbs update.

The Acceptance-Rate Dial

Efficiency lives in the middle

step too small $\Rightarrow$ accept ≈ 0.90 (crawls); too large $\Rightarrow 0.06$ (stuck); sweet spot $\approx 0.23\text{--}0.44$

Both extremes fail

- High acceptance: steps too small to explore.
- Low acceptance: chain rejects and sticks.
- Either way, the samples are correlated.

The rule of thumb

- Aim near 0.44 in one dimension,
- and near 0.234 in many dimensions.
- That maximizes effective samples.

Three Step Sizes

Read the acceptance rate

step size	acceptance	behavior
small (0.5)	0.90	tiny steps, high autocorrelation
good (2.5)	0.35	near the 0.23-0.44 sweet spot
large (8.0)	0.06	mostly rejected, chain stuck

The small step accepts almost everything but crawls; the large one rejects almost everything and sticks. Only the middle step - about a third acceptance - mixes efficiently.

The Gibbs Alternative

Draw each conditional, no rejection

Gibbs: draw each conditional in turn $\theta_1 \sim p(\theta_1 \mid \theta_2, D)$, $\theta_2 \sim p(\theta_2 \mid \theta_1, D)$ (no rejection)

How it works

- Update one parameter at a time.
- Draw it from its conditional, holding others fixed.
- Every draw is accepted - no tuning.

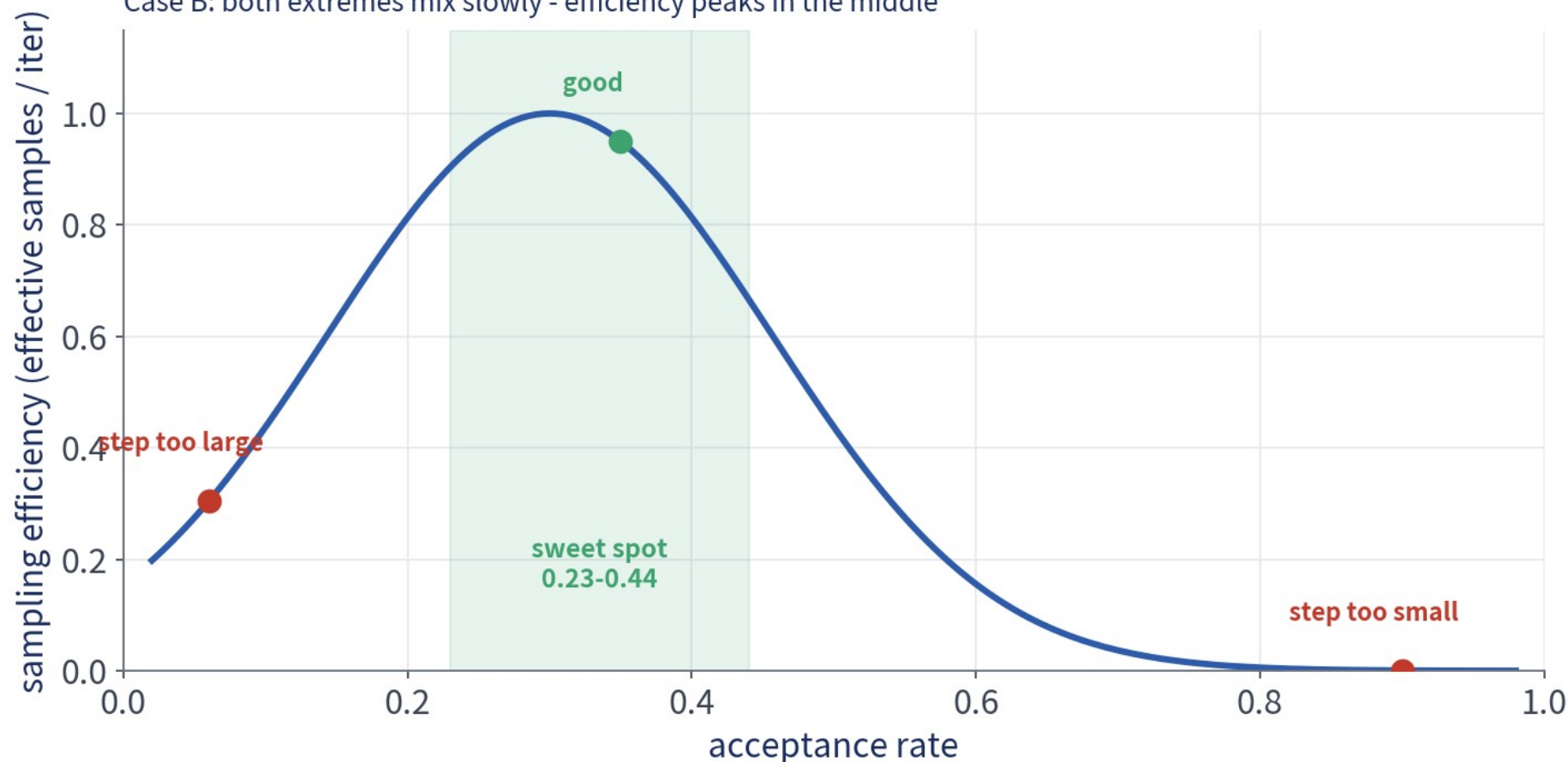
The catch

- You need the conditionals in closed form.
- Great for conjugate models,
- harder when they don't factor nicely.

Efficiency Peaks in the Middle

Sampling efficiency - effective samples per iteration - is low at both extremes and peaks in the shaded 0.23-to-0.44 band. A 0.90 acceptance means tiny useless steps; a 0.06 acceptance means a stuck chain. The good step size sits right at the top of the hump

Case B: both extremes mix slowly - efficiency peaks in the middle



Two Ways to the Same Chain

Tune, or draw exactly

Metropolis-Hastings

- Works almost anywhere - just need the target.
- But you must tune the proposal.
- The acceptance rate is your guide.

Gibbs

- No tuning, no rejection.
- But needs closed-form conditionals.
- Ideal for conjugate structure.

Build the Chain Two Ways

Propose-and-reject chain - ~~Both samplers construct a chain whose long-run law is the posterior. Metropolis-Hastings proposes and rejects, using only the target ratio - so the evidence cancels - but needs its step size tuned to an efficient acceptance rate. Gibbs sidesteps rejection by drawing each conditional exactly, when you can. Same destination, two~~

The step (Case A)

- One ratio decides each move.
- The evidence cancels out.
- Worse moves are taken for exploration.

The tuning (Case B)

- Acceptance rate sets efficiency.
- Aim for the 0.23-0.44 band.
- Or use Gibbs and skip it.

Reproduce the Step and the Tuning

Direct Claude Code to check your arithmetic

GOAL Verify one MH step, the acceptance-rate tuning, and a Gibbs update.

[INPUT] Prompt · Claude Code

```
Write and run part2_practice.py that:  
CASE A - one MH step:  
  * current target 0.30; score proposals 0.42 and  
    0.18; print each ratio and the accept decision  
    (use u=0.45 for the downhill move).  
CASE B - tuning:  
  * run MH on a target at step sizes 0.5, 2.5, 8.0;  
    print the empirical acceptance rate for each;  
  * also code a 2-parameter Gibbs sampler.  
Then plot the trace and the efficiency vs acceptance.  
Match every printed number to the slides.
```

Check against the slides

- uphill ratio 1.40 -> accept.
- downhill 0.60, u=0.45 -> accept.
- accept rates 0.90, 0.35, 0.06.
- Gibbs: acceptance = 1.

Deliverable: part2_practice.py whose printed numbers match the MH step and the acceptance rates.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 Why does the evidence cancel in the ratio?

It appears in both posteriors, top and bottom

Q2 Why accept some worse moves?

To explore the whole posterior, not just the peak

Q3 When would you use Gibbs over Metropolis?

When the conditionals are available in closed form

All three come from the MH acceptance step and the tuning-versus-Gibbs choice.

Three Pitfalls to Avoid

Where the samplers trip people

Common mistakes

- "A high acceptance rate means good sampling."
- "Compute the evidence to score a move."
- "Gibbs always beats Metropolis."

The reality

- Too high means tiny steps - it barely explores.
- The evidence cancels; only the ratio matters.
- Gibbs needs closed-form conditionals to work.

Two Calculations to Remember

Part 2 applications in one screen

1

One MH step is a ratio and a coin flip

From a target of 0.30, the uphill move to 11% scored 1.40 and was accepted; the downhill move to 4% scored 0.60 and was accepted anyway when a uniform draw of 0.45 fell below it. The evidence cancelled in both ratios.

2

Tune the proposal by its acceptance rate

A tiny step accepted 90% but crawled; a large one accepted 6% and stuck; the good step accepted about a third - near the 0.23-0.44 sweet spot where efficiency peaks. Gibbs skips this by drawing each conditional exactly.

3

Two routes, one chain

Metropolis-Hastings proposes and rejects using only the target ratio, tuned to a good acceptance rate; Gibbs draws conditionals with no rejection. Both build a chain whose long-run law is the posterior.

Next — Part 3 practice: Hamiltonian Monte Carlo and NUTS, which propose smarter, farther moves.